A* Examination Guide

Built from 17 question papers and 16 mark schemes, January 2019 to June 2026

This guide is built from the Paper 2H question papers and mark schemes. Methods are laid out the way the mark schemes award M, A and B marks, and every warning comes from a mark-scheme note such as "working required", "do not ISW", "SC" or "correct answer from obvious incorrect working scores 0". The formulae sheet is printed in the paper; everything else must be learnt.

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2 Twenty mark-scheme rules
3 Number: powers, standard form, surds, bounds
4 Percentages, ratio and proportion
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Papers analysed: Jan 2019 (question paper only), Jun 2019, Jan 2020, Nov 2020, Jan 2021, Jun 2021, Nov 2021, Jan 2022, Jun 2022, Jan 2023, Jun 2023, Nov 2023, Jun 2024, Nov 2024, Jun 2025, Nov 2025, Jun 2026. Papers are cited by series, e.g. (Jun 25).

1 HOW PAPER 2H IS BUILT

Format: 2 hours, 100 marks, 21–26 questions, calculator allowed. Paper 2H covers the same specification as Paper 1H with no topic restrictions, but certain question types recur here more often: **ruler-and-compass constructions**, **algebraic proof**, **recurring decimals to fractions**, **matching graphs to equations**, **solving by drawing a straight line on a graph**, **three-set Venn diagrams with conditional probability**, **comparing distributions** and **trigonometric graphs y = a sin x + b**.

Topic	Papers (of 17)	What is asked most
Percentages, ratio, compound change	17	reverse two-stage change, profit, depreciation, compare % increases
Indices, expanding, factorising, subject	17	triple brackets, subject appears twice, fractional indices
Statistics (CF, histograms, mean, IQR)	16	CF graph, histogram probability, combined mean, compare two groups
Probability (trees, without replacement)	15	"at least one", conditional from Venn or histogram, unknown x
Sets and Venn diagrams	13	3-set Venn from statements, n(A ∩ B′), probability given a set
Trigonometry (2D and 3D)	14	sine/cosine rule, ambiguous obtuse case, angle with a plane, bearings
Recurring decimals	11	"use algebra to show that 0.37878… = 25/66"
Quadratic inequalities	10	factorise, critical values, region from a sketch
Non-linear simultaneous equations	10	line with circle $x^2 + y^2 = r^2$ or a quadratic curve
Functions	10	composite, inverse (including restricted domain), excluded value

Topic	Papers (of 17)	What is asked most
Circle theorems	10	find angles with reasons; intersecting chords; inscribed circles
Bounds	10	upper/lower bound of a formula; suitable degree of accuracy
Vectors	9	find a ratio or prove a trapezium using a vector method
Graphs (match, draw a line, transformations)	9	graph shapes, y = a sin x + b, tangent to estimate gradient
Algebraic proof	4	consecutive squares, odd numbers, remainders
Constructions	3	perpendicular bisector, angle bisector (show all arcs)

2 TWENTY MARK-SCHEME RULES

1. **Working is required** wherever the question says "show clear algebraic working", "use algebra" or "show that": an answer alone scores 0.
2. **No working and a wrong answer scores 0;** a correct answer from obviously incorrect working also scores 0.
3. **"Show that" questions:** finish exactly at the given result and include every intermediate line.
4. **Recurring decimals:** both multiplied equations (e.g. 1000x and 10x) and the subtraction must be seen; quoting a rule scores nothing.
5. **Algebraic proof:** use general terms (2n + 1 for odd, n and n + 1 for consecutive), expand fully, and write a concluding statement ("8(…) + 2, so the remainder is 2"). Numerical examples score 0.
6. **Constructions:** leave all construction arcs; no arcs, no marks.
7. **Do not ISW a wrong simplification** of a correct answer; stop when the answer is correct.
8. **Multipliers:** 0.85 or 1 − 0.15 is accepted; "(1 − 15%)" is not.
9. **Proportion:** write the formula with k ($Q = k\sqrt{d}$, $M = kh^3$) before substituting.
10. **Venn diagrams:** fill from the centre outwards; every region including "none" must be correct.
11. **Conditional probability:** the denominator is the given group only ("given that this student chose photography").
12. **Comparisons of distributions:** one comparison of an average (median) and one of spread (IQR), each in context ("on average Office A workers travel further").
13. **Graph solutions:** draw the line you use; give solutions to the accuracy asked (usually 1 d.p.).
14. **Gradient estimates:** draw a tangent of at least 3 cm and use its rise over run.
15. **Bounds:** show the bounds used; "suitable degree of accuracy" needs both bounds and a reason.
16. **Angles:** give each circle-theorem reason in full; "because of the circle theorem" is not enough.
17. **Quadratic inequalities:** sketch or test; give both regions with correct signs ($x < a$ or $x > b$; $a \le x \le b$).
18. **Vectors:** use a vector method (two routes and equated coefficients); for a trapezium show one pair of sides parallel by a scalar multiple.
19. **Accuracy:** 3 s.f. unless told otherwise; keep full values in the calculator.
20. **Answer line:** the final answer must be on the answer line; incorrect values in working with no answer line lose the A mark.

3 NUMBER: POWERS, STANDARD FORM, SURDS, BOUNDS

RECURRING DECIMAL METHOD (2–3 MARKS)

Let $x = 0.5393939\ldots$; $1000x = 539.393939\ldots$; $10x = 5.393939\ldots$; subtract: $990x = 534$; $x = 534/990 = 89/165$ (Jun 26).

Same pattern: $0.3787\ldots = 25/66$ (Nov 25); $0.6818\ldots = 15/22$ (Nov 20). Choose powers of 10 so the recurring parts line up exactly.

- Index laws, negative and fractional powers: $(16y^{10}/(9x^4))^{-½} = 3x^2/(4y^5)$; write each number as a prime power to find n in $18 \times 27^{4n+6} = 3^x$ type questions.
- Standard form calculations: give the answer as $a \times 10^n$, $1 \le a < 10$.
- Surds: $\sqrt{8} = 2\sqrt{2}$; $(5 - \sqrt{18})/(1 - \sqrt{2})$: multiply by $(1 + \sqrt{2})$; show every stage. Surd lengths (Jun 23): solve $x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5}$ by rationalising.
- LCM/HCF from prime powers: LCM takes the highest power of each prime, HCF the lowest.
- Bounds (Jun 23): density $= 3M/(4r^3)$; upper bound uses $M = 265$ and $r = 2.75 \Rightarrow 9.56$ g/cm^3.

4 PERCENTAGES, RATIO AND RATES

WORKED EXAMPLES

Reverse two-stage (Jan 19): +15% then −8% gives 687 700 ⇒ original = 687 700 ÷ (1.15 × 0.92) = 650 000.

Compare increases (Jan 20): 500 → 545 is 9%; 545 → 592 is 8.62%, so the later increase is smaller.

Discount from the sale price (Jan 20): 952 = 0.85 × normal ⇒ normal 1120 ⇒ discount 168 euros.

Overall change: +15% then −8% ⇒ 1.15 × 0.92 = 1.058, a 5.8% increase (Jun 19).

- Speed conversions: × 1000 ÷ 3600; 50 m/s = 180 km/h; x km/h = 5x/18 m/s.
- Density, pressure: $\rho = m/V$, $p = F/A$; volume conversions 1 m^3 = 1 000 000 cm^3.
- Value for money (Jan 20): convert to the same currency and unit price before comparing.
- Ratio and fractions of groups (Jan 21): turn each part into a fraction of the whole, then add.

5 ALGEBRA AND PROOF

ALGEBRAIC PROOF (3–4 MARKS)

Jan 20: $(n + 1)^2 - n^2 = 2n + 1$, which is odd for any integer n.

Jan 22: $(2n + 1)^2 + (2n + 3)^2 = 8n^2 + 16n + 10 = 8(n^2 + 2n + 1) + 2$, so the remainder on division by 8 is 2.

- Subject appearing twice: $p = (ac + 8)/(3 + c) \Rightarrow 3p + pc = ac + 8 \Rightarrow c(p - a) = 8 - 3p \Rightarrow c = (8 - 3p)/(p - a)$.
- Fractions in equations: multiply every term by the LCM of the denominators first.
- Algebraic fractions: factorise fully before cancelling; answers like $(x - 7)/(x + 5)$ must be fully simplified.
- Completing the square: $3x^2 - 6x + 5 = 3(x - 1)^2 + 2$; $28 + 24x - 6x^2 = 52 - 6(x - 2)^2$, turning point (2, 52).
- Quadratic formula given in a form (Nov 25): match $(-4 \pm 2\sqrt{39})/10$ to $(-b \pm \sqrt{(b^2 - 4ac)})/2a$ to find a and c.

6 EQUATIONS AND INEQUALITIES

NON-LINEAR SIMULTANEOUS EQUATIONS · JAN 23

$y = 7 - 2x$ and $x^2 + y^2 = 34 \Rightarrow 5x^2 - 28x + 15 = 0 \Rightarrow (5x - 3)(x - 5) = 0 \Rightarrow (5, -3)$ and (0.6, 5.8).

Midpoint of PQ (Nov 21): solve, then average the coordinates.

- Quadratic inequality (Jun 24): $3y^2 + 4y - 32 > 0 \Rightarrow (3y - 8)(y + 4) > 0 \Rightarrow y < -4$ or $y > 8/3$.

- Inequalities from areas (Nov 20, Jan 22): form the quadratic, solve, then apply the context (lengths > 0).
- Linear programming regions: draw x = 1.5, y = x, x + y = 6 accurately; label region R.
- Simultaneous linear equations: eliminate, substitute back, give both values.

7 SEQUENCES AND FUNCTIONS

- Arithmetic series (Nov 25): a = 10, $u_{20} = 86 \Rightarrow d = 4$; $S_N = N/2[20 + 4(N - 1)] = 5194 \Rightarrow 2N^2 + 8N - 5194 = 0 \Rightarrow N = 49$.
- Series with algebraic terms (Nov 21): find d from consecutive terms, then use the given term to find k.
- Composite gf(x): g of f; simplify. Inverse: swap and rearrange.
- Restricted domain (Jan 22): $g(x) = 5 + 6x - x^2 = 14 - (x - 3)^2$, $x \geq 3 \Rightarrow g^{-1}(x) = 3 + \sqrt{(14 - x)}$, domain $x \leq 14$.
- $f^{-1}(7)$ directly: solve f(x) = 7.

8 COORDINATE GEOMETRY AND GRAPHS

GRAPH SKILLS SPECIFIC TO PAPER 2

Match graphs: $y = x^2$ (U), $y = -1/3\ x^2$ (∩), $y = -5/x$ (reciprocal in 2nd/4th quadrants), cubic, y = sin x, exponential.

Solve using a drawn graph: rearrange the new equation so one side is the plotted function; the other side is the straight line to draw. With $y = 2x^2 - 4x - 1$ plotted, $x^2 - x - 1 = 0 \Rightarrow 2x^2 - 4x - 1 = -2x + 1$, so draw $y = -2x + 1$ (Jan 22).

Gradient at a point: draw a tangent and calculate rise ÷ run (Jun 26).

Trig graphs: y = a sin x + b has amplitude a and centre line y = b; y = a cos(x − b)° + c: read the maximum, minimum and the shift.

- Perpendicular lines: $m_1 m_2 = -1$. 5y + 2x = 7 has gradient −2/5, so the perpendicular has gradient 5/2.
- Kites and rhombuses (Jun 22, Jun 23): diagonals are perpendicular; the longer one bisects the shorter.
- Transformations: f(x + a), f(x) + b, af(x), f(ax), −f(x); describe fully ("reflection in the x-axis").

9 CALCULUS

WORKED EXAMPLES

Nov 25: $y = x^3 - 16x + 7$, gradient $11 \Rightarrow 3x^2 - 16 = 11 \Rightarrow x = \pm 3$; tangents $y = 11x - 47$ and $y = 11x + 61$, so b = −47 or 61.

Jan 22: $x = 4t^3 - 27t + 8$; direction reverses when $v = 12t^2 - 27 = 0 \Rightarrow t = 1.5$; $a = 24t = 36$ m/s^2.

Turning points (Jun 25): dy/dx = 0, solve the quadratic, substitute for y.

10 GEOMETRY AND TRIGONOMETRY

CIRCLE THEOREMS AND CHORDS

Centre angle = 2 × circumference angle · semicircle 90° · same segment equal · cyclic quadrilateral 180° · tangent ⊥ radius · equal tangents · alternate segment.

Intersecting chords (Jan 22): PT × TQ = ST × TR ⇒ 3PT = 48 ⇒ PT = 16 ⇒ diameter 19 ⇒ radius 9.5 cm.

Inscribed circle in an equilateral triangle (Nov 23, Nov 24): the centre is where the angle bisectors meet; radius = side ÷ (2√3).

- Ambiguous sine rule (Jan 22): if the angle is obtuse, use 180° minus the calculator value.

- Area of a quadrilateral (Nov 21): split into two triangles, use the cosine rule for the diagonal and ½ab sin C.
- 3D angles (Jun 23, Jun 25): find the foot of the perpendicular, build the right-angled triangle, use tan.
- Bearings (Nov 24): back bearing = bearing ± 180°; use the cosine rule for AC, then the sine rule for the angle.
- Constructions: perpendicular bisector (arcs from both ends, equal radius) and angle bisector (arc on both arms, then equal arcs); keep all arcs.

11 MENSURATION AND SIMILARITY

- Sphere from surface area (Jan 23): $4\pi r^2 = 49\pi \Rightarrow r = 3.5 \Rightarrow V = 180$ cm^3.
- Similar solids (Jan 23): area ratio 25/64 ⇒ length 5/8 ⇒ volume 125/512; $V_A - V_B = 541.8 \Rightarrow V_A = 716.8$, $V_B = 175$ cm^3.
- Nov 25: surface areas 200 and 450 ⇒ scale 2 : 3 ⇒ volume $672 \times 27/8 = 2268$ cm^3.
- Frustum (Nov 23): volume = big cone − small similar cone; set up the equation in x.
- Composite shapes with shading (Jun 22): area of square − circles = 40; solve for the side.
- Show-that surface area (Jun 25): use the volume to express y in terms of x, then substitute into A.

12 VECTORS

METHOD

Express the vector to the unknown point two ways with scalars, equate the a and b coefficients, solve.

Trapezium proof (Nov 20): with OC : CA = OD : DB = 1 : 2, CD = (1/3)(b − a) = (1/3)AB, so CD is parallel to AB and ABDC is a trapezium.

Magnitude: $|(x, y)| = \sqrt{(x^2 + y^2)}$; add column vectors component by component.

13 STATISTICS

- Combined means: total = mean × n. Sixth item (Jan 21): 6 × 401 − 5 × 398 = 416 g. Girls' mean from class and boys' totals (Jun 19).
- CF graph: median at n/2, quartiles at n/4 and 3n/4; "more than 48 minutes" = total − reading at 48.
- Histograms: frequency = density × width; partial bars use a fraction of the width; conditional (Jan 20): "given more than 30 minutes" uses only that part as the denominator.
- Compare two groups: "The median for A is higher, so on average…"; "The IQR for B is smaller, so B's results are more consistent." Use the actual values.
- IQR from a list: order first; LQ at the (n + 1)/4th value.

14 PROBABILITY AND SETS

WORKED EXAMPLES

At least one red still in the bag (Jun 23): 3 red, 9 green, three counters taken. P(all three reds taken) = (3/12)(2/11)(1/10) = 1/220, so the answer is 219/220.

With replacement (Jun 26): P(at least one red) = 39/64 ⇒ P(no red) = 25/64 ⇒ P(green) = 5/8 ⇒ P(exactly one red) = 2 × 3/8 × 5/8 = 15/32.

Three-set Venn (Jun 22, Jun 24, Jun 25): start with all three, then each pair minus the centre, then singles, then use the total to find "none" or x.

- Conditional from a Venn diagram: P(goats | chickens) = n(G ∩ C) ÷ n(C).
- Set notation: n(A ∩ C′) counts in A but not C; (A ∪ B)′ is outside both.

- Trees: complete all branches (pairs sum to 1); “at least one” = 1 − P(none).

15 FINAL A* CHECKLIST

1. Read for “show that”, “use algebra”, “show clear algebraic working”: every line must be written.
2. Recurring decimals: two equations lined up, subtract, simplify.
3. Proofs: general algebraic terms, full expansion, concluding sentence.
4. Constructions: arcs visible. Graph solutions: line drawn.
5. Venn diagrams: centre first; conditional probabilities use the given set as the denominator.
6. Comparisons: median (average) and IQR (consistency), in context, with values.
7. Percentages: multipliers; reverse changes divide by the product of multipliers.
8. Inequalities and simultaneous equations: factorise, critical values, correct regions, paired solutions.
9. Bounds and accuracy: correct bound choices; 3 s.f. unless told otherwise.
10. Circle theorems: reasons in words. Leave 10 minutes to recheck answers and units.